

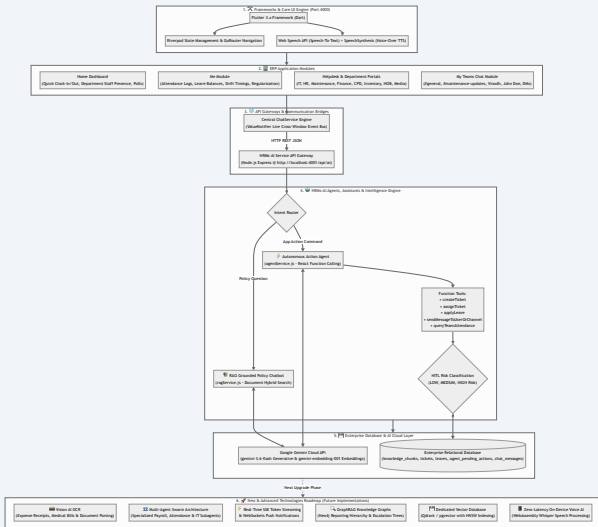
HRMs-ERP & HRMs-AI Master Architecture Blueprint

Enterprise System Design, Subsystem Specifications & RAG Knowledge Ingestion

v2.5.0 Production

Active Sync

1. Master System Architecture Chart



2. Feeding Knowledge to the AI Agent (RAG Pipeline)

To ensure the AI Agent answers company questions accurately without hallucinations, raw enterprise documents are processed through a 5-step Knowledge Feeding & Grounding Pipeline:

1

Document Collection & Extraction

Raw text is collected from company source files including HR Leave Manuals, Employee Handbooks, IT/AV Equipment SOPs, Benefits Guides, and Resolved Ticket Histories.

2

Semantic Chunking & Metadata Tagging

Documents are split into optimal 500-1000 character overlapping text chunks. Each chunk is tagged with structured metadata (category, department, target_role, updated_timestamp).

3

High-Dimensional Vector Embedding

Each chunk is processed through the Google Gemini Embedding API (gemini-embedding-001) to convert human language into a 3072-dimensional vector math representation.

4

Dual Indexing (Vector & Full-Text)

Chunks, metadata, and 3072-vector embeddings are stored in the `knowledge_chunks` table, indexing both BM25 Full-Text keywords and vector cosine similarity distance.

5

Real-Time Hybrid Search & Grounded Synthesis

When an employee asks a question, the query is embedded and matched via Hybrid Search (Cosine Similarity + BM25). The top K matched contexts are injected into Gemini system prompts to generate 100% accurate, policy-grounded answers.

3. Core Layer Specifications

**Layer 1: Core UI & Voice Engine**

- **Frontend Stack:** Flutter 3.x Web & Mobile built with Dart.
- **State Management:** Riverpod providers combined with custom `ValueNotifier` event bus.
- **Voice System:** HTML5 Web Speech API for Mic Speech-To-Text paired with Web SpeechSynthesis API for Voice-Over TTS audio output.

**Layer 2: ERP Application Modules**

- **Home Dashboard:** Web clock-in/out, staff presence modal, announcements, polls.
- **Me Section:** Monthly attendance, work modes, leave balances, regularizations.
- **Helpdesk:** Service tickets across IT, HR, Maintenance, Finance, CPD, Inventory, HOB, Media.
- **Teams Chat:** Channels (`#general`, `#maintenance-updates`) and DMs.

**Layer 3: API Gateways & Synchronization**

- **Central ChatSync:** ChatService event bus routes messages from floating chatbots and AI drawers to target team chats live.
- **Express Backend API:** Service at `http://localhost:4001/api/ai` exposing `/chat`, `/agent`, `/confirm`, `/decline`.

**Layer 4: HRMs-AI Engine & HITL Safety**

- **RAG Policy Chatbot:** Hybrid BM25 FullText + Cosine Similarity vector search over 3072-dim embeddings.
- **Action Agent:** ReAct function tool calling (`createTicket`, `assignTicket`, `applyLeave`, `sendMessageToUserOrChannel`, `queryTeamAttendance`).
- **HITL Risk Evaluation:** Staged in `agent_pending_actions` table with interactive UI confirmation cards.

4. Advanced Technology Roadmap

Technology	Capability	Implementation Goal
 Vision AI OCR	Multimodal Vision Analysis	Auto-extract expense receipt line items, medical bills, and ID proofs into reimbursement tickets.
 Multi-Agent Swarm	Subagent Orchestration	Decouple AI into specialized subagents (*Payroll Agent*, *Attendance Audit Agent*, *IT Agent*) via Redis Pub/Sub.
 Real-Time SSE Streaming	Token-by-Token Delivery	Stream AI answers letter-by-letter with typing indicators and WebSockets push alerts.
 GraphRAG Knowledge Graphs	Hierarchy Graph Traversal	Integrate Neo4j to query organizational reporting structures and escalation trees.
 Dedicated Vector DB	HNSW Indexing	Migrate vector storage to Qdrant or pgvector for sub-millisecond similarity search across millions of chunks.
 On-Device Voice AI	Quantized SLM Execution	Embed Whisper models via WebAssembly inside browser for zero-latency offline speech recognition.

Note: This architecture document is synced with the production codebase. Open this HTML document directly in Chrome or Safari to view the live interactive Mermaid flowchart diagram and generate print-ready PDF reports.